



ADAPTING TO EXTREMES

The lesser flamingo has evolved to occupy a very specialized niche: to feed on algae in extremely alkaline African lakes. In the absence of competitors, it can thrive there in huge numbers.

ISLAND EVOLUTION

Isolated islands provide natural laboratories for unusually fast evolution, with intense competition for limited resources leading to rapid speciation. In 1835, Darwin's visit to the Galapagos Islands allowed him to collect many bird specimens, particularly finches. He noted slight variations between the specimens from island to island. He also heard about the differences between the giant tortoises on separate islands, and subsequent visits to other Pacific islands made him wonder about the possibility of new species evolving from a common ancestor. The ornithologist John Gould was able to identify Darwin's finches as a new group of 12 separate species, rather than just varieties of the same species.

This persuaded Darwin that species could change under certain conditions such as island isolation. Island wildlife continues to be an important research area for modern evolutionary biologists.



< FLIGHTLESS BIRDS

Many species of flightless birds, such as kiwis, evolved on the islands of New Zealand, which lacked serious predators until humans arrived.

ARTIFICIAL SELECTION

Over millennia, humans have domesticated many different kinds of animals and plants, from dogs and cattle to fruit trees and cereal crops. Before the discovery of the gene, this was achieved simply by selectively interbreeding organisms bearing the desired characteristics – such as the ability to run fast or produce more succulent fruits – over many generations, until the selected traits became dominant. Today, biotechnology achieves the same result much faster by directly manipulating genes, both to enhance beneficial traits and to remove problematic ones.



^ GENETIC MODIFICATION

Alteration to the genetic makeup of organisms can remove undesired characteristics and add more useful ones, such as resistance to disease.



^ CLONING

Genetically identical individuals can be created by the transfer of a nucleus – with its genetic information – from an adult cell into a host egg cell.